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Carbon cycling by arbuscular mycorrhizal fungi in soil–plant systems

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Arbuscular mycorrhizal fungi (AMF) play an important role in regulating carbon fluxes between the biosphere and the atmosphere. A recent study showed that live hyphae can turn over rapidly, in five to six days on average, suggesting that carbon flow to AMF hyphae might be respired back to the atmosphere quickly. However, that study gives a limited view of the residence time of AMF hyphae in soils. AMF hyphae can also contribute to soil carbon storage through other mechanisms.

The soil organic carbon (SOC) pool, an important component of terrestrial ecosystems, is a crucial regulator of carbon fluxes between the biosphere and the atmosphere. Mechanisms influencing SOC storage depend mainly on net primary production and the distribution of photosynthate between above- and below-ground structures. Although primary production is a major determinant in the sequestration of carbon in soils, it is the size and activity of the microbial biomass of the soil that regulates carbon accumulation via mineralization and immobilization of plant and microbially derived residues in the soil. The exact amount of sequestration appears to depend on land management practices, edaphic factors, climate, and the amount and quality of plant and microbial inputs. The sequestration of carbon in soils used for agriculture, forestry, and land reclamation has been recognized as a potential option to mitigate global change [1–4]. Recent research suggests that mycorrhizal fungi might be an important component of the SOC pool, in addition to facilitating carbon sequestration by stabilizing soil aggregates.

AMF and carbon fluxes between the biosphere and the atmosphere

Symbiotic associations between plant roots and arbuscular mycorrhizal fungi (AMF) are ubiquitous in terrestrial ecosystems [5]. The role of AMF in mediating the ecosystem response to global change has been reviewed recently [6,7]. In view of the importance of AMF in ecosystem processes, these reviews highlight a significant void in research addressing the contributions of AMF to terrestrial carbon cycling. Recent studies using ¹⁴C labeling indicated that photosynthate is transferred from host plants to AMF hyphae within hours after labeling [8]. It is also generally accepted that AMF receive all their carbohydrate from the host plant and that the association of AMF with roots could create a sink demand for carbohydrate (increasing sink strength), which could result in a 4–20% drain of carbon from the host plant and could indirectly influence carbon storage in soils [9]. Furthermore, up-regulation of photosynthesis by AMF is indicated where the amount of fungus in the root system is related directly to net carbon gain of the host [10]. Such fungus-mediated effects on plant growth can potentially improve carbon sequestration by increasing net primary production, especially in nutrient-limited environments.

External hyphae and carbon storage in soils

The AMF could directly influence soil carbon dynamics through the growth and turnover of extra-radical hyphae (ERH) within bulk soil. The residence time of ERH in soil is usually thought to be short (days), and it has rarely been measured explicitly. Philip Staddon *et al.* recently used accelerator mass spectrometry microanalysis of ¹⁴C to quantify the turnover rate of ERH in plants grown in a

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controlled environment [11]. A surprising outcome of this study is that the turnover rate of ERH attached to plant roots averaged five to six days. The authors indicated that carbon flow from host plants to AMF in soil might quickly be respired back to the atmosphere. More importantly, their findings suggest a rapid pathway for atmospheric carbon to enter the soil carbon cycle. Because AMF hyphal cell walls are composed primarily of chitin – a carbohydrate that is rather recalcitrant to decomposition – the rapid turnover of live ERH would still allow for the accumulation of hyphal residues that could remain within the soil matrix for a considerable period of time. Currently, little information is available on the residence time of chitinous cell wall residues, particularly in a soil matrix, although recent studies using pyrolysis GC/MS-C-IRMS (gas chromatography-mass spectrometry-combustion interface-isotope ratio-mass spectrometry) indicate a residence time of 49 ± 19 years for protein, amino acid or chitin-derived pyrolysis products [12]. Measurement of hyphal residue accumulation would have been difficult in the short time period of the Staddon *et al.* [11] experiment. Furthermore, the use of potting medium consisting of sand and attapulgite clay eliminated physical protection – a major mechanism for stabilizing hyphal residues exposed to soil microbial activity (and hence for increasing residence time in soil) – from the experiment. Moreover, the Staddon *et al.* [11] study did not determine whether the hyphae (characterized by rapid turnover) were decomposed completely to CO₂ or remained as residues within the potting medium.

The typical dry weight of ERH in soil, $\sim 0.03\text{--}0.5\text{ mg g}^{-1}$, represents a large proportion of soil microbial biomass [13,14]. For a soil depth of 30 cm with bulk density of 1.2 g cm^{-3} and 50% carbon content of dry hyphae, the amount of SOC derived directly from AMF ranges from 54 to 900 kg ha⁻¹. This range of ERH values indicates that in spite of the rapid turnover of live hyphae, the amount of carbon retained by ERH in the soil is measurable, and the maintenance of a stable hyphal network is functionally important for the sequestration of carbon below-ground. The rather high turnover values reported by Staddon *et al.*, in combination with results demonstrating a rather large stock of ERH biomass, suggest that more than one pool of ERH exists, probably distinguished by hyphal architecture [11,15]. One pool is composed of hyphae with relatively fast turnover (days), probably related to the hyphal architectural type known as exploratory or absorptive hyphae. Another pool with relatively slower turnover (weeks) is composed of the thicker walled ERH with arbuscular architecture. These observations suggest a need for future research to consider ERH turnover and the resultant contribution to carbon sequestration owing to hyphal architecture.

ERH and soil aggregation

If the turnover values reported by Staddon *et al.* [11] can be generalized to all ERH, we might have to re-evaluate the contributions of AMF to soil structure. However, the studies to date indicate that ERH are relatively persistent in soil. Indeed, the influences of ERH on soil aggregation might be even more important to the carbon stock than the

influence of the hyphal standing crop alone [16]. Through their role in soil macroaggregate stabilization, the ERH appear to contribute to the creation of an aggregate hierarchy. In doing so, they help to create a mechanism for increasing the residence time of organic debris within soil macroaggregates. Mechanistically, the ERH contribution to soil aggregation can be viewed as a 'sticky-string-bag' mechanism, in which the hyphae help to entangle and enmesh soil particles to form macroaggregate structures [16,17]. The physical dimensions of the ERH allow them to grow and to ramify through soil pores the size of those between macroaggregates. A basic premise of the sticky-string-bag mechanism is the existence of a relatively stable hyphal phase, with its filamentous nature, coarse branching habit, and rather large diameters.

The contribution of AMF hyphae to carbon cycling lies not only with ERH themselves, but also with exudates from hyphae. We know that AMF hyphae are responsible for the production of a glycoprotein-like substance, glomalin [18], which is fairly stable in soils [19]. Radiocarbon dating of the operationally defined glomalin extract indicates a residence time in soils of 6–42 years [20], much longer than the residence time reported for AMF hyphae. In a tropical forest soil, glomalin carbon was shown to represent up to $\sim 4\text{--}5\%$ of total soil carbon, much higher than soil microbial biomass carbon [13]. The close correlation of the amount of glomalin in soil with hyphal length and the stability of soil aggregates [21] suggests that glomalin could influence soil carbon storage indirectly by stabilizing soil aggregates. One of the modes of action of glomalin could be facilitating the formation of a sticky string bag of hyphae, the primary mode by which AMF contribute to soil aggregation [8]. However, with glomalin still only operationally defined, the challenges ahead are to isolate and characterize the substance, to identify the mechanism of its stabilizing action in soils, and to define the relationship

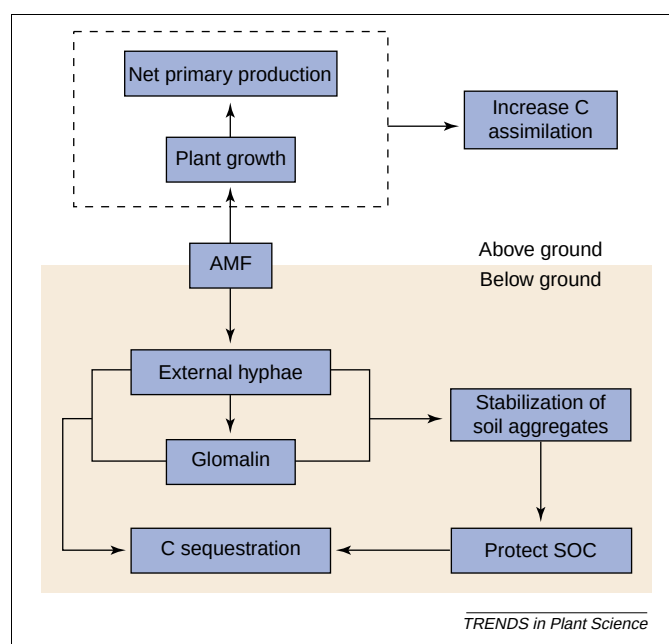


Fig. 1. Role played by arbuscular mycorrhizal fungi (AMF) in regulating carbon fluxes between the biosphere and the atmosphere. Abbreviation: SOC, soil organic carbon.

between hyphal turnover and glomalin inputs to the soil. Only then can the relative contribution of glomalin to carbon cycling be determined.

Concluding remarks

The AMF can influence carbon fluxes between the biosphere and the atmosphere through different pathways (Fig. 1). A key AMF-mediated process involved in the storage of carbon in soils is the transfer of photosynthate from host plants to AMF hyphae. Although the turnover of external hyphae linked to plant roots appears to be rapid, the overall contribution of AMF to soil carbon storage could depend significantly on the kinds of hyphae produced, the residence time of accumulated hyphal residues, the production of glomalin, and the role played by AMF in the stabilization of soil aggregates.

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Spontaneous and thermoinduced photon emission: new methods to detect and quantify oxidative stress in plants

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Peroxidation of polyunsaturated fatty acids is one of the main events triggered by oxidative stress in cells. Some lipid peroxidation products are light-emitting species, and their luminescence can be used as an internal marker of oxidative stress. However, this spontaneous chemiluminescence is weak and difficult to measure. Recent studies have shown that an alternative

approach that involves measuring thermoluminescence bands at high temperature (in the range 80–150°C) is a simple way of detecting and quantifying lipid peroxidative damage and oxidative stress in plants.

Oxygen is essential for the metabolism of aerobic organisms. However, its participation in metabolic reactions leads inevitably to the production of partially reduced forms of oxygen (superoxide, hydrogen peroxide, hydroxyl

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